Summary

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Infinite Limits

$\lim\limits_{x \to a} f(x) = \infty$ means the values of $f(x)$ grow arbitrarily large as x approaches a.

$\lim\limits_{x \to a} f(x) = -\infty$ means the values of $|f(x)|$ grow arbitrarily large as x approaches a with $f(x)$ negative.

Limits at Infinity

$\lim\limits_{x \to \infty} f(x) = L$ means the values of $f(x)$ becomes arbitrarily close to L by making x sufficiently large.

$\lim\limits_{x \to -\infty} f(x) = L$ means the values of $f(x)$ becomes arbitrarily close to L by making $|x|$ sufficiently large with x negative.

Vertical and Horizontal Asymptotes:

A function f has a **vertical asymptote** at $x = a$ if at least one of the following conditions hold:

- $\lim\limits_{x \to a} f(x) = \pm\infty$
- $\lim\limits_{x \to a^+} f(x) = \pm\infty$
- $\lim\limits_{x \to a^-} f(x) = \pm\infty$

A function f has a **horizontal asymptote** at $y = L$ if at least one of the following conditions hold:

- $\lim\limits_{x \to \infty} f(x) = L$
- $\lim\limits_{x \to -\infty} f(x) = L$

Both vertical asymptotes and horizontal asymptotes are written as the equation of a line. Vertical asymptotes are given as the equation of a vertical line, and horizontal asymptotes are given as the equation of a horizontal line.